RAYMOND (RAY) YOUNG

he/him/his
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RESEARCH STATEMENT

My research focuses on current-aware autonomy/optimal path planning, primarily in simulation where we can analyze a large number of paths for data-driven insights. More generally, I enjoy applying theory from robotics, optimization, and estimation to improve the autonomous capabilities of marine vehicles.

EDUCATION

Scripps Institution of Oceanography 2020-present
Ph.D. Candidate - Expected Graduation 11/2025

- Major: Oceanography (Applied Ocean Science)
- Advisor: Dr. Sophia Merrifield
- Lab: Coastal Observing Research and Development Center (CORDC)

University of California San Diego

- M.S., Mechanical Engineering 2022
- B.S., Mechanical Engineering (*Cum Laude*) 2020

HONORS AND AWARDS

- National Defense Science and Engineering Graduate (NDSEG) Fellowship 2022-2025
- Scripps Fellowship Fall 2020

IN REVIEW/IN PREP

1. **R Young**, FH Akins, M Anderson, S Merrifield. *Energy-Optimal Path Planning for Autonomous Underwater Vehicles via Imitation Learning and Transformers.* (in prep)
2. **R Young**, FH Akins, S Merrifield. *Energy-Optimal Path Planning for Autonomous Underwater Vehicles with the Maximum Principle*, IEEE Journal of Oceanic Engineering. (in review)

JOURNAL PUBLICATIONS

1. M Siegelman, W ORielly, J Becker, C Olfe, **R Young**, PL Colin, E Terrill, S Merrifield. *Spectral Refraction Modeling of Waves around the Steep Reef at Palau*, Journal of Geophysical Research: Oceans. (2025) (accepted)
2. **R Young**, S Merrifield, M Anderson, M Mazloff, E Terrill. *A Greedy Depth Seeking Behavior for Energy-Efficient Transits by an Autonomous Underwater Vehicle,* IEEE Journal of Oceanic Engineering. (2024) DOI
3. S Merrifield, S Celona, RA McCarthy, A Pietruszka, H Batchelor, R Hess, A Nager, **R Young**, K Sadorf, LA Levin, DL Valentine, JE Conrad, EJ Terrill. *Wide-Area Debris Field and Seabed Characterization of a Deep Ocean Dump Site Surveyed by Autonomous Underwater Vehicles*, Environmental Sciences and Technology. (2023) DOI

CONFERENCE PUBLICATIONS

1. M Anderson, A Nager, S Merrifield, E Terrill, **R Young**, J Chang. *Analysis and Test of an Optimal Propeller Speed Schedule to Reduce AUV Energy Consumption*, in proceedings of OCEANS 2025: Brest, Brest, FRA. (2025) DOI
2. T Kwan*, **R Young**, A Nager, M Anderson, E Terrill, S Merrifield. *Chasing Currents: Onboard Depth Optimization for AUV Energy Savings*, in proceedings of OCEANS 2024: Halifax, Halifax, CAN. (2024) DOI

3. **R Young***, S Merrifield, M Anderson, E Terrill. *A Simulation Framework for Environmentally-Aware Autonomous Underwater Vehicle (AUV) Mission Planning and Algorithm Development*, in proceedings of OCEANS 2023-MTS-IEEE US Gulf Coast. (2023) DOI
4. J Chang*, M Anderson, S Merrifield, A Nager, R Hess, **R Young**, S Kitchen, and E Terrill. *Power Efficiency Autonomy for Long Duration AUV Operation*, in proceedings of the 2022 IEEE/OES Autonomous Underwater Vehicles Symposium (AUV), Singapore. (2022) DOI
5. **R Young***, S Merrifield, M Anderson, E Terrill. *USV Navigation Geometry for Surface Current Inference,* in proceedings of OCEANS 2021: San Diego-Porto, San Diego, USA. (2021) DOI
6. A Amador*, S Merrifield, R McCarthy, **R Young**, E Terrill. *Wave Glider Speed for Real-time Planning,* in proceedings of OCEANS 2021: San Diego-Porto, San Diego, USA. (2021) DOI
7. S Merrifield, J Shapiro, **R Young***, LS Laurent, H Simmons, M Otero, ... E Terrill. *Autonomy system for USV/UUV coordinated sampling,* in proceedings of OCEANS 2019 MTS/IEEE Seattle, Seattle, USA. (2019) DOI

** presenting author*

SELECTED PRESENTATIONS

1. **R Young**, FH Akins, S Merrifield. *Time-Optimal Path Planning in 3-D with the Maximum Principle*, 2024 IEEE OES AUV Symposium, Boston, MA. (2024) (poster)
2. **R Young**. *Current-Aware Autonomy*, 5th Annual NDSEG Fellowship Conference, New Orleans, LA. (2024) (oral/poster)
3. **R Young**, FH Akins, S Merrifield. *Optimal Path Planning for Underwater Vehicles: Methods and Simulation*, SIO Applied Ocean Science Seminar, La Jolla, CA. (2024) (oral)
4. **R Young**, FH Akins, S Merrifield. *Time Optimal Path Planning for Underwater Vehicles in Time-Dependent Flows with the Maximum Principle*, Southern California Applied Mathematics Symposium, La Jolla, CA. (2024) (poster)
5. **R Young**, S Merrifield, M Anderson, E Terrill. *A Sampling Based Approach to Energy-Optimal Planning in Time-Varying Flows*, 2022 IEEE/OES AUV Symposium, Singapore. (2022) (oral)
6. **R Young**, S Merrifield, M Anderson, E Terrill. *Current Aware Planning for Long-Duration USVs in a Time-Varying Environment*, 2022 Ocean Sciences Meeting, Virtual. (2022) (oral)

FIELD WORK

ONR DRI First- and Second Island Arc Turbulent Eddy Regional Exchange (ARCTERX)

- TN441 - 12 days. R/V Thomas G. Thompson - Guam, USA 2025
- TN417 - 16 days. R/V Thomas G. Thompson - Korror, Palau 2024
- RR2203 - 12 days. R/V Roger Revelle - Guam, USA 2023
- Role: Scientist - Data analysis of shipboard data: ADCP, Coherent Doppler Radar, and flow-through CTD. Deployment, recovery, and data analysis for automomous vehicles: Wave Glider USVs; Slocom and and Remus AUVs.

Mapping and Water Column Exploration Offshore Palau

- NA169 - 17 days. E/V Nautilus - Koror, Palau 2024
- Role: Scientist - Data analysis of shipboard data: ADCP, echosounder, and flow-through CTD. Deployment, recovery, and data analysis for autonomous vehicles: Wave glider and Wam-V USVs.
- Expedition Summary

ONR DRI Northern Ocean Rapid Surface Evolution (NORSE)

- AR60-02 - 35 days. R/V Neil Armstrong - Reykjavik, Iceland 2023
- Role: Scientist - assisted deployment/recovery and operation of a Wave Glider for collaborative sampling.

TEACHING

Instructor of Record
UCSD, MAE Department

- MAE 142: Dynamics and Control of Aerospace Vehicles — Summer 2025

Teaching Assistant
UCSD, SIO Department

- SIO 111: Introduction to Ocean Waves, prof. Falk Fedderson — Winter 2024
- SIOC 209: Hacking for the Oceans, prof. Sophia Merrifield — Spring 2021

Volunteer Instructor
SIO Incoming Graduate Student Mathematics Workshop

- Lectures on Probability and Statistics — Fall 2022, 2024

Training
UCSD Teaching and Learning Commons

- Advanced College Teaching: Equitable Course Design & Instruction — 2025
- Introduction to College Teaching: Foundations of Equitable Teaching — 2024

MENTORING/SERVICE

Mentoring

- Erin Beazley - Electrical Engineering Georgia Tech (with S Merrifield) — Summer 2025
- Trevor Kwan - Computer Engineering UCSD (with S Merrifield) — 2024
- Luke Colosi & Oceane Boulais - Scripps Graduate Student Peer Mentor Program — 2021-2023

Service

- Incoming Graduate Student Mathematics Workshop (leadership) — 2024
- Scripps Graduate Student Peer Mentor Program (leadership) — 2022-2023
- Applied Ocean Science Seminar Organizing Committee — 2021-2022
- Unlearning Racism in Geosciences (URGE) subpod member on demographic data — 2021-2022

PROFESSIONAL EXPERIENCE

ASML/Cymer Light Source — Summer 2019
Manufacturing Engineering Intern

- Automated transfer of information from existing to new Manufacturing Execution System (MES) with scripting in C# and VBA (Visual Basics)
- Attended daily meetings with technicians on the manufacturing floor
- Calibrated electronic testing systems
- Provided documentation for mechanical and electrical, build and test procedures

Coastal Observing Research and Development Center (CORDC) — 2018 - 2020
Undergraduate Researcher/Engineering Intern

- Analyzed platform motion and sensor performance from a Boeing Liquid Robotics Wave Glider, an unmanned surface vehicle (USV)
- Developed algorithms for adaptive path planning behaviors in Python, MATLAB
- Designed sensor mounts in SolidWorks
- Fieldwork experience deploying and piloting USVs from SIO pier and R/V Robert Gordon Sproul